

# ZAYAAN BHANWADIA

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## EDUCATION

### University of Toronto

Honours B.Sc. in Computer Science (Co-op) | GPA: 3.7/4.0

Toronto, ON

Sept. 2025 – Apr. 2029 (Expected)

## EXPERIENCE

### Full-Stack Developer Intern

KorraNet Creative Inc. (Riipen)

May 2026 – Present

Winnipeg, Manitoba

- Improved FastAPI backend's reliability and security by restructuring environment variable ingestion and removing hardcoded secrets.
- Implemented Meta OAuth token handling by identifying gaps in token storage, expiration, and implementing fixes for better integration of AI models.
- Added API error handling and timeout prevention to increase backend and system architecture stability to reduce system crashes and failed API calls.
- Implemented logging across onboarding and authentication flows to isolate bugs and support new developer teams.

### Co-President

NASA Hunch Program

Sept. 2023 – June 2025

Toronto, Ontario

- Secured a 3rd-place international finish as the executive of the first Canadian team to compete in NASA Hunch.
- Engineered an autonomous Lunar Robot using Arduino, and CAD simulation to execute navigation on lunar terrain.
- Programmed a real-time C++ control algorithm utilizing non-blocking execution loops to prevent system unresponsiveness and synchronize timed image capture sequences.

## PROJECTS

### Embedded Key-Value Store | C++, POSIX Threads, Linux

- Built an embedded C++ database using data sharding to prevent global lock bottlenecks and improve database throughput.
- Used shared locks to enable non-blocking concurrent reads across threads while maintaining thread-safe writes.
- Implemented data persistence by writing updates to a log file on disk before updating memory to handle crashes.
- Created a file compaction routine to safely clean up old or deleted keys and reduce disk usage without pausing the app.

### Context Sync Extension | TypeScript, VS Code Extension API

- Developed a VS Code extension with over **500 downloads** that automates AI chat context sharing to maintain workspace continuity across developer environments.
- Designed a structured Markdown format utilizing weighted graph edges to optimize the token-usage-to-context ratio.
- Integrated GitHub Copilot, OneDrive, and Google Drive to automate syncing and provide easy integration.
- Developed a lightweight Node.js backend to handle secure, fast data synchronization and state management.

### Clinical AI Assistant (Top 10 Projects @ GenAI Genesis) | Python, SQL, OpenCV

- Developed an AI clinical assistant using IBM Watsonx to automate patient check-ins and structure collected data.
- Engineered a live monitoring dashboard with a computer vision module for real-time fall detection.
- Developed a spatial visualization tool that generates 3D labelled hospital models by extruding floor plans.

### Automated Table Tennis Referee (Finalist @ Hack Canada) | Python, C, YOLO

- Engineered a live computer vision system using OpenCV and YOLO coordinates to track and automate scoring game.
- Developed game-state logic and scoring algorithms by processing coordinates from a YOLO object detection.

## TECHNICAL SKILLS

**Languages:** Java, Python, C, C++, SQL, JavaScript, Typescript, Bash, HTML/CSS

**Libraries/Frameworks:** OpenCV, NumPy, Flask, FastAPI, Node.js, React

**Developer Tools:** Linux, Git, Github, Docker, PGAdmin, VS Code, Jira/Confluence, Slack

**Cloud/Databases:** PostgreSQL, IBM Cloud, Google Cloud Project, Vercel, Vultr, Railway, Supabase